

ISAAC ROSENBERG

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EDUCATION

• Erasmus Mundus Joint Master in Astrophysics

Master of Science in Astrophysics

September 2025 – July 2027

Italy, Serbia, France

- Two-year joint program across University of Rome Tor Vergata, University of Belgrade and Université Côte d'Azur focused on astrophysics and space science.

- **Current grade:** 29.7 / 30

• University of Toronto

Honours Bachelor's of Science

September 2021 – May 2025

Toronto, Canada

- Astronomy and Astrophysics Specialist, Mathematics Minor.

- **Grade:** 3.85 / 4.00

PUBLICATIONS

1. **Isaac Rosenberg**, Martine Lokken, Renée Hložek, Weiguang Cui, Sara Santoni, Romeel Davé (2026). [Here, There and Everywhere: How AGN Jets Affect Galaxy Cluster Environments](#). Submitted for publication in *Astronomy & Astrophysics*.

RESEARCH EXPERIENCE

The Impacts of AGN Feedback on Galaxy Cluster Environments

2024 - 2026

Department of Astronomy, University of Toronto

- Research project with Professor Renée Hložek and Dr. Martine Lokken.
- Ran hydrodynamic simulations with the 300 collaboration of galaxy clusters to understand how AGN feedback affects galaxy and cluster properties.
- Paper submitted to *Astronomy & Astrophysics*.

Standard Siren Cosmology with Gravitational Waves

2024 - 2026

Canadian Institute for Theoretical Astrophysics

- Research project with Professor Maya Fishbach.
- Built a model that can simultaneously fit for the cosmological and astrophysical evolution of the binary black hole mass distribution.
- Received Summer Undergraduate Research Fellowship to continue this project over summer 2025.
- Paper in preparation.

Summer Research, ATLAS Experiment

Summer 2024

CERN, Switzerland

- Worked under Professors Nikolina Ilic and Richard Teuscher on the ATLAS experiment at CERN for 12 weeks, in person.
- Investigated yearly spatial variations in jet distributions of the $H \rightarrow ZZ$ particle decay.

CONFERENCES, SCHOOLS & TALKS

1. The Three Hundred Annual Collaboration Meeting

August 2026

La Cristalera, Spain

Talk

Research talk on AGN feedback simulations; five days of discussions and workshops on galaxy cluster simulations.

2. International Conference on Physics Students

August 2026

Rabat, Morocco

Talk

Research talk on gravitational wave cosmology; five days of lectures and workshops across broad topics in physics.

3. Flatiron Institute CMB Summer School

July 2026

New York, US

School

Five-day workshop on CMB data analysis with the Simons Observatory.

4. Master in Astrophysics and Space Science Winter School

Feb 2026

Rome, Italy

School

Four-day workshop on machine learning and data science applications to astrophysics.

5. FLATs Talk Series <i>University College London, UK</i> Research talk on galaxy clusters.	January 2026 Talk
6. Cosmology Meeting <i>Imperial College London, UK</i> Research talk on galaxy clusters.	January 2026 Seminar
7. Simulations Meeting <i>University of Oxford, UK</i> Research talk on galaxy clusters.	January 2026 Seminar
8. CMB – Large Scale Structure Meeting <i>University of Cambridge, UK</i> Research talk on galaxy clusters.	January 2026 Seminar
9. EXPLORE <i>Toronto, Canada</i> Research talk on gravitational wave cosmology; winner of best talk award.	August 2025 Talk
10. CASCA <i>Halifax, Canada</i> Research poster and talk on galaxy clusters.	June 2025 Poster
11. STATSTRO <i>Toronto, Canada</i> Research talk on gravitational wave cosmology.	May 2025 Talk

HONORS AND AWARDS

• Erasmus Mundus Joint Masters Scholarship <i>Erasmus MASS program</i>	2025 - 2027 €34,000
• Winner of Best Talk Award at EXPLORE Summer School <i>York University</i>	2025 [🏆]
• Summer Undergraduate Research Fellowship <i>Canadian Institute for Theoretical Astrophysics</i> ◦ Received \$9,500 to continue my research on gravitational waves.	Summer, 2025 \$9,500 [🌐]
• Regents Admission Scholarship <i>Victoria College, University of Toronto</i>	2021, \$5,000
• Regents In-course Scholarship I <i>Victoria College, University of Toronto</i>	2022, \$1,000
• Prof. William Kingston and Dr. John Kingston Scholarship <i>Victoria College, University of Toronto</i>	2023, \$1,000
• John Hamilton Watson Award for International Study <i>Victoria College, University of Toronto</i>	2024, \$3,000
• CIE International Experience Award <i>Centre for International Experience, University of Toronto</i>	2024, \$1,500
• Helen Mae Woodliffe Scholarship <i>Victoria College, University of Toronto</i>	2024, \$1,000

TEACHING EXPERIENCE

Teaching Assistant <i>University of Toronto, Department of Mathematics & Department of Astronomy</i>	2023 - 2025
• Teaching assistant for MAT186 (Calculus I) , MAT187 (Calculus II) , MAT188 (Linear Algebra) , and AST251 (Life on Other Worlds) .	
• Led three tutorials per week, marked assignments, invigilated exams, and held office hours for university calculus, linear algebra, and astronomy courses with over 4,000 students combined.	

RELEVANT COURSEWORK

* Graduate-level courses in *italics*

Astrophysics

- Stars and Planets, Galaxies and Cosmology, Introduction to Astrophysics, Astrophysics Laboratory, *Cosmology, Stellar Astrophysics, Radiative Processes, Active Galactic Nuclei, Plasma Spectroscopy, Computational Astrobiology, Interstellar Medium, General Relativity.*

Physics

- Classical Mechanics I & II, Electromagnetism I & II, Thermal Physics, Computational Physics, Physics Laboratory, *High Energy Physics, Quantum Mechanics.*

Mathematics and Computer Science

- Analysis, Multivariable Calculus, Linear Algebra I & II, Differential Equations, Introduction to Computer Programming, Neural Networks, *Complex Analysis, ODEs & PDEs.*

SKILLS

- **Programming Languages:** Python, C, C++, Perl, Java, MATLAB, ROOT
- **Hydrodynamic Simulations (and related tools):** GIZMO, GADGET, CAESAR, YT, DisPerSE
- **Python Packages:** Numpy, Scipy, Matplot, Pandas, Astropy, PyTorch, scikit-learn

PROFESSIONAL MEMBERSHIPS

- **The Three Hundred Collaboration** *October, 2024 - Present*

ADDITIONAL INFORMATION

Languages: English (native), French (intermediate), Russian (intermediate), German (beginner)
Interests: Guitar, singing, world history, baseball, travel, chess, politics, music, football.